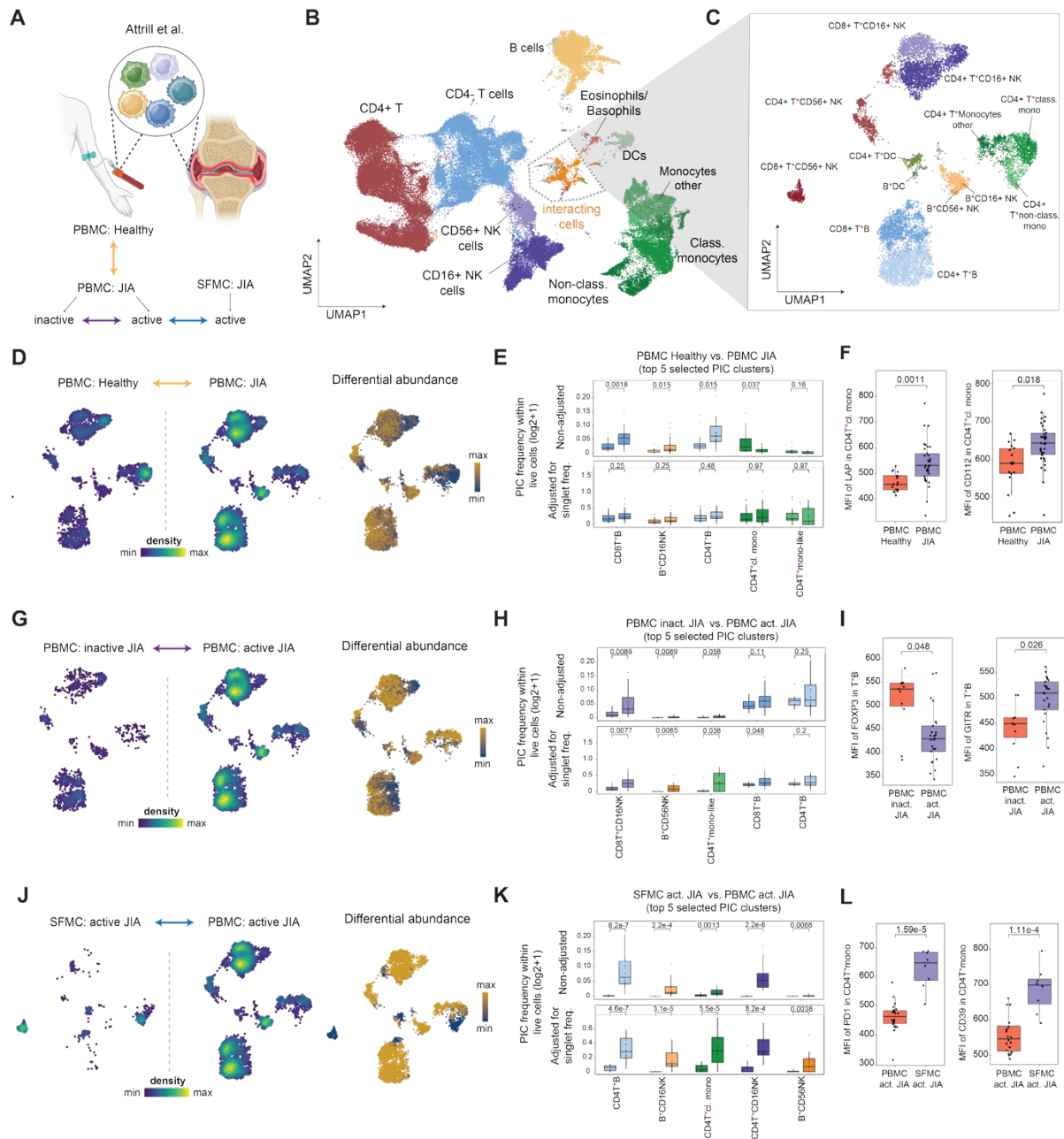




Interacting cell landscape in juvenile idiopathic arthritis (JIA). **A.** Publicly available spectral flow cytometry data of PBMCs and SFMCs of JIA patients. Three comparisons (indicated by the arrows) were made for the interacting cell landscape. **B.** UMAP of the overall cellular landscape. Recorded cells were processed with PICtR, out of 7,843,646 cells, 80,000 sketched cells are displayed. **C.** UMAP of interacting cells ($n = 12,908$). **D.** Point density UMAP (left panel) and differential abundance (right panel) of interacting cells comparing PBMCs from healthy donors vs. JIA patients. **E.** Quantitative comparisons of interacting cell frequencies between PBMCs from healthy donors ($n=18$) and JIA patients ($n=36$). Top: Non-adjusted frequencies. Bottom: Interaction frequencies adjusted by the harmonic mean of the singlet frequencies of the contributing cells (see Methods). P values were determined with a two-sided t-test and adjusted for multiple testing using Benjamini-Hochberg correction. **F.** Qualitative differences in CD4T*cl.mono interactions. P values were determined with a two-sided Wilcoxon rank sum test and adjusted for multiple testing using Benjamini-Hochberg correction. **G.** Point density UMAP (left panel) and differential abundance (right panel) of interacting cells comparing PBMCs of JIA patients with inactive disease ($n=11$) vs. active ($n=25$). **H.** Quantitative comparisons of interacting cell frequencies between PBMCs from JIA with inactive and active disease. Top: Non-adjusted frequencies. Bottom: Interaction

frequencies adjusted by the harmonic mean of the singlet frequencies of the contributing cells (see Methods). P values were determined with a two-sided t-test and adjusted for multiple testing using Benjamini-Hochberg correction. **I.** Qualitative differences in T*B interactions. P values were determined with a two-sided Wilcoxon rank sum test and adjusted for multiple testing using Benjamini-Hochberg correction. **J.** Point density UMAP (left panel) and differential abundance (right panel) of interacting cells comparing PBMCs of JIA patients with active disease vs. SFMC of active disease. **K.** Quantitative comparisons of interacting cell frequencies between PBMCs of JIA patients with active disease (n=25) vs. SFMC of active disease (n=8). Top: Non-adjusted frequencies. Bottom: Interaction frequencies adjusted by the harmonic mean of the singlet frequencies of the contributing cells (see Methods). P values were determined with a two-sided t-test and adjusted for multiple testing using Benjamini-Hochberg correction. **L.** Qualitative differences in CD4T*mono interactions. P values were determined with a two-sided Wilcoxon rank sum test and adjusted for multiple testing using Benjamini-Hochberg correction. Abbreviations: UMAP = uniform manifold approximation and projection, PBMC = peripheral blood mononuclear cells, SFMC = synovial fluid mononuclear cells. Red asterisks in cell type labels indicate interactions between the respective cell types. Box plots display the median, first and third quartiles and whiskers are defined as 1.5 times interquartile range.